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Hasan Sözer

Curriculum Vitæ

Last updated on July 29, 2018

Education

2009 **Ph.D. in Computer Science**, *University of Twente*, Enschede, The Netherlands.
Advisors: Dr. Bedir Tekinerdoğan and Dr. Mehmet Akşit
Thesis title: "Architecting Fault-Tolerant Software Systems"

2004 **M.Sc. in Computer Engineering**, *Bilkent University*, Ankara, Turkey.
Advisor: Dr. İbrahim Körpeoğlu
Thesis title: "A Peer-to-Peer File Sharing System for Wireless Ad-Hoc Networks"

2002 **B.Sc. in Computer Engineering**, *Bilkent University*, Ankara, Turkey.

Employment

2017 – present **Associate Professor**, *Ozyegin University*, İstanbul, Turkey.
Department of Computer Science

2011 – 2017 **Assistant Professor**, *Ozyegin University*, İstanbul, Turkey.
Department of Computer Science

2009 – 2011 **Post-doctoral Researcher**, *University of Twente*, Enschede, The Netherlands.
Faculty of Electrical Engineering, Mathematics and Computer Science

2005 – 2009 **Ph.D. Associate**, *University of Twente*, Enschede, The Netherlands.
Faculty of Electrical Engineering, Mathematics and Computer Science

2002 – 2005 **Software Engineer**, *Aselsan Inc.*, Ankara, Turkey.
Radar, Electronic Warfare and Intelligence Systems Division

Research

Interests: Software Engineering, Software Architecture Design, Software Testing, Software Fault Tolerance, Self-adaptive and Distributed Systems.

Research Projects

2014 – 2017 Automated Model Refinement for Model-based Testing
Principal Investigator, Özyeğin University, İstanbul, Turkey
Carrying industrial partner: Vestel Electronics
Funded by the Turkish Ministry of Science, Industry and Technology

2014 – 2017 Automated Integration of Static and Dynamic Software Analysis Techniques
Principal Investigator, Özyeğin University, İstanbul, Turkey
Funded by the Scientific and Technological Research Council of Turkey

2012 – 2014 Fault Handling and Quality Optimization in Service-Oriented Systems
Principal Investigator, Özyeğin University, İstanbul, Turkey
Carrying industrial partner: Vestel Electronics
Funded by the Turkish Ministry of Science, Industry and Technology

2012 – 2014 Crew Pairing and Fleet Assignment Models for Turkish Airlines
Researcher, Özyeğin University, İstanbul, Turkey
Funded by the Scientific and Technological Research Council of Turkey

2009 – 2011 Octopus: Smart System Adaptability
Researcher, Embedded Systems Institute, Eindhoven, The Netherlands
Carrying industrial partner: Canon Nederland
Funded by the Netherlands Ministry of Economic Affairs

2005 – 2009 Trader: Reliability by Design
Researcher, Embedded Systems Institute, Eindhoven, The Netherlands
Carrying industrial partner: NXP
Funded by the Netherlands Ministry of Economic Affairs

2002 – 2005 Bluetooth Scatternet Construction and Bluetooth Appications
Researcher, Bilkent University, Ankara, Turkey
Funded by the Scientific and Technological Research Council of Turkey

Publications

Journal Articles

[1] C.S. Gebizli, A. Kirkici, and H. Sozer. Increasing test efficiency by risk-driven model-based testing. *Journal of Systems and Software*, 144:356–365, 2018.

[2] M.F. Kirac, B. Aktemur, and H. Sozer. VISOR: A fast image processing pipeline with scaling and translation invariance for test oracle automation of visual output systems. *Journal of Systems and Software*, 136:266–277, 2018.

[3] H. Sozer and C.S. Gebizli. Model-based testing of Digital TVs: An industry-as-laboratory approach. *Software Quality Journal*, 25(4):1185–1202, 2017.

[4] C.S. Gebizli and H. Sozer. Automated refinement of models for model-based testing using exploratory testing. *Software Quality Journal*, 25(3):979–1005, 2017.

[5] O.O. Ozener, M. Ormeci Matoglu, G. Erdogan, M. Haouari, and H. Sozer. Solving a large-scale integrated fleet assignment and crew pairing problem. *Annals of Operations Research*, 253(1):477–500, 2017.

[6] H. Sozer, M. Stoelinga, H. Boudali, and M. Aksit. Availability analysis of software architecture decomposition alternatives for local recovery. *Software Quality Journal*, 25(2):553–579, 2017.

[7] G. Karagoz and H. Sozer. Reproducing failures based on semi-formal failure scenario descriptions. *Software Quality Journal*, 25(1):111–129, 2017.

[8] T. Ayav and H. Sozer. Identifying critical architectural components with spectral analysis of fault trees. *Applied Soft Computing*, 49:1270–1282, 2016.

[9] H. Sozer. Integrated static code analysis and runtime verification. *Software: Practice and Experience*, 45(10):1359–1373, 2014.

[10] K. Gulcu, H. Sozer, B. Aktemur, and A.O. Ercan. Fault masking as a service. *Software: Practice and Experience*, 44(7):838–854, 2014.

[11] A. de Roo, H. Sozer, and M. Aksit. Composing domain-specific physical models with general-purpose software modules in embedded control software. *Software and Systems Modeling*, 13(1):55–81, 2014.

[12] A. de Roo, H. Sozer, L. Bergmans, and M. Aksit. MOO: An architectural framework for runtime optimization of multiple system objectives in embedded control software. *Journal of Systems and Software*, 86(10):2502–2519, 2013.

[13] H. Sozer, B. Tekinerdogan, and M. Aksit. Optimizing decomposition of software architecture for local recovery. *Software Quality Journal*, 21(2):203–240, 2013.

[14] B. Tekinerdogan, H. Sozer, and M. Aksit. Feature-based rationale management system for supporting software architecture adaptation. *International Journal of Software Engineering and Knowledge Engineering*, 22(7):945–964, 2012.

[15] A. de Roo, H. Sozer, and M. Aksit. Verification and analysis of domain-specific models of physical characteristics in embedded control software. *Information and Software Technology*, 54(12):1432–1453, 2012.

[16] H. Sozer, B. Tekinerdogan, and M. Aksit. FLORA: A framework for decomposing software architecture to introduce local recovery. *Software: Practice and Experience*, 39(10):869–889, 2009.

[17] H. Sozer, M. Tekkalmaz, and I. Korpeoglu. A peer-to-peer file search and download protocol for wireless ad hoc networks. *Computer Communications*, 32(1):41–50, 2009.

[18] B. Tekinerdogan, H. Sozer, and M. Aksit. Software architecture reliability analysis using failure scenarios. *Journal of Systems and Software*, 81(4):558–575, 2008.

[19] M. Tekkalmaz, H. Sozer, and I. Korpeoglu. Distributed construction and maintenance of bandwidth and energy efficient bluetooth scatternets. *IEEE Transactions on Parallel and Distributed Systems*, 17(9):963–974, 2006.

Book Chapters

[1] B. Tekinerdogan and H. Sozer. An architecture viewpoint for modeling dynamically configurable software systems. In I. Mistrik, N. Ali, R. Kazman, J. Grundy, and B. Schmerl, editors, *Managing Trade-offs in Adaptable Software Architectures*, pages 79–97. Elsevier, Amsterdam, The Netherlands, 2016.

[2] H. Sozer, B. Tekinerdogan, and M. Aksit. Software architecture design for local recovery. In R. Mathijssen, editor, *Trader-Reliability of High-Volume Consumer Products*, pages 139–150. Embedded Systems Institute, Eindhoven, The Netherlands, 2009.

[3] H. Sozer, B. Tekinerdogan, and M. Aksit. Extending failure modes and effects analysis approach for reliability analysis at the software architecture level. In R. de Lemos, C. Gacek, and A. Romanovsky, editors, *Architecting Dependable Systems IV*, volume 4615 of *Lecture Notes in Computer Science*, pages 409–433. Springer, 2007.

Publications in Refereed Conferences and Workshops

[1] A. Kirkici, C. Sahin Gebizli, and H. Sozer. Risk-driven model-based testing of washing machine software: An industrial case study. In *Proceedings of the 13th Workshop on Testing: Academic and Industrial Conference – Practice and Research Techniques*, pages 398–403, Vasteras, Sweden, 2018.

[2] M. Altinisik, E. Ersoy, and H. Sozer. Evaluating software architecture erosion for PL/SQL programs. In *Proceedings of the 4th Workshop on Software Architecture Erosion and Architectural Consistency*, pages 159–165, Canterbury, United Kingdom, 2017.

[3] B. Ergun, C. Sahin Gebizli, and H. Sozer. FORMAT: A tool for adapting test models based on feature models. In *Proceedings of the 9th IEEE International Workshop on Software Test Automation*, pages 66–71, Turin, Italy, 2017.

[4] K. Incki, I. Ari, and H. Sozer. Runtime verification of IoT systems using complex event processing. In *Proceedings of the 14th IEEE International Conference on Networking, Sensing and Control*, pages 625–630, Calabria, Italy, 2017.

[5] I. Yildirim and H. Sozer. Automated extraction of feature models from android based portable devices. In *Proceedings of the 2017 IEEE International Workshop on Software Engineering and Knowledge Management*, pages 441–448, Prague, Czech Republic, 2017.

[6] Y. Kilic and H. Sozer. Generating runtime verification specifications based on static code analysis alerts. In *Proceedings of the 32nd ACM Symposium on Applied Computing*, pages 1342–1347, Marrakech, Morocco, 2017.

[7] C. Sahin Gebizli and H. Sozer. Impact of education and experience level on the effectiveness of exploratory testing: An industrial case study. In *Proceedings of the 12th Workshop on Testing: Academic and Industrial Conference – Practice and Research Techniques*, pages 23–28, Tokyo, Japan, 2017.

[8] E. Ersoy, K. Kaya, M. Altinisik, and H. Sozer. Using hypergraph clustering for software architecture reconstruction of data-tier software. In *Proceedings of the 10th European Conference on Software Architecture*, pages 326–333, Copenhagen, Denmark, 2016.

[9] E. Ersoy and H. Sozer. Extending static code analysis with application-specific rules by analyzing runtime execution traces. In *Proceedings of the 31st International Symposium on Computer and Information Sciences*, pages 30–38, Krakow, Poland, 2016.

[10] C. Sahin Gebizli and H. Sozer. Model-based software product line testing by coupling feature models with hierarchical markov chain usage models. In *Proceedings of the 6th IEEE International Workshop on Model-Based Verification and Validation*, pages 278–283, Vienna, Austria, 2016.

[11] T. Ayav and H. Sozer. Moving switching functions to continuous domain. In *Proceedings of the 6th IEEE International Workshop on Model-Based Verification and Validation*, pages 265–270, Vienna, Austria, 2016.

[12] C. Sahin Gebizli, H. Sozer, and A.O. Ercan. Successive refinement of models for model-based testing to increase system test effectiveness. In *Proceedings of the 11th Workshop on Testing: Academic and Industrial Conference – Practice and Research Techniques*, pages 263–268, Chicago, IL, USA, 2016.

[13] C. Sahin Gebizli, D. Metin, and H. Sozer. Combining model-based and risk-based testing for effective test case generation. In *Proceedings of the 10th Workshop on Testing: Academic and Industrial Conference – Practice and Research Techniques*, pages 1–4, Graz, Austria, 2015.

[14] M. Altinisik and H. Sozer. Automated procedure clustering for reverse engineering PL/SQL programs. In *Proceedings of the 31st ACM Symposium on Applied Computing*, pages 1440–1445, Pisa, Italy, 2016.

[15] F.Ozturk, E.Sarili, H.Sozer, and B.Aktemur. Effort estimation for architectural refactoring to introduce module isolation. In *Proceedings of the 8th European Conference on Software Architecture*, pages 300–307, Vienna, Austria, 2014.

[16] C. Sahin Gebizli and H. Sozer. Improving models for model-based testing based on exploratory testing. In *Proceedings of the 6th IEEE International Workshop on Software Test Automation*, pages 656–661, Vasteras, Sweden, 2014.

[17] U. Yuksel, H. Sozer, and M. Sensoy. Trust-based fusion of classifiers for static code analysis. In *Proceedings of the 17th International Conference on Information Fusion*, pages 0–0, Salamanca, Spain, 2014.

[18] F. Koca, H. Sozer, and R. Abreu. Spectrum-based fault localization for diagnosing concurrency faults. In *Proceedings of the 25th International Conference on Testing Software and Systems*, pages 239–254, Istanbul, Turkey, 2013.

[19] H. Sozer, S. Nouta, A. Wombacher, and P. Perona. DAPNA: An architectural framework for data processing networks. In *Proceedings of the 28th International Symposium on Computer and Information Sciences*, pages 221–229, Paris, France, 2013.

[20] U. Yuksel and H. Sozer. Automated classification of static code analysis tool alerts: A case study. In *Proceedings of the 29th IEEE International Conference on Software Maintenance*, pages 532–535, Eindhoven, The Netherlands, 2013.

[21] S. Ciraci, H. Sozer, and B. Tekinerdogan. A run-time verification framework for smart grid applications implemented on simulation frameworks. In *Proceedings of the 2nd International Workshop on Software Engineering Challenges for the Smart Grid*, pages 1–8, San Francisco, CA, USA, 2013.

[22] K. Gulcu, H. Sozer, and B. Aktemur. FAS: Introducing a service for avoiding faults in composite services. In *Proceedings of the 4th International Workshop on Software Engineering for Resilient Systems*, pages 106–120, Pisa, Italy, 2012.

[23] M. Astekin and H. Sozer. Utilizing clone detection for domain analysis of simulation systems. In *Proceedings of the 10th Working IEEE/IFIP Conference on Software Architecture*, pages 287–291, Helsinki, Finland, 2012.

[24] B. Tekinerdogan and H. Sozer. Variability viewpoint for introducing variability in software architecture viewpoints. In *Proceedings of the 2nd International Workshop on Variability in Software Architecture*, pages 163–166, Helsinki, Finland, 2012.

[25] S. Ciraci, H. Sozer, and B. Tekinerdogan. An approach for detecting inconsistencies between behavioral models of the software architecture and the code. In *Proceedings of the 36th IEEE International Conference on Computer Software and Applications*, pages 257–266, Izmir, Turkey, 2012.

[26] K. Incki, I. Ari, and H. Sozer. A survey of software testing in the cloud. In *Proceedings of the 6th IEEE International Conference on Software Security and Reliability*, pages 18–24, Washington, D.C., USA, 2012.

[27] A. Yildiz, B. Aktemur, and H. Sozer. Rumadai: A plug-in to record and replay client-side events of web sites with dynamic content. In *Proceedings of the 2nd Workshop on Developing Tools as Plug-ins*, pages 88–89, Zurich, Switzerland, 2012.

[28] H. Sozer, C. Hofmann, B. Tekinerdogan, and M. Aksit. Runtime verification of component-based embedded systems. In *Proceedings of the 26th International Symposium on Computer and Information Sciences*, pages 471–477, London, UK, 2011.

[29] B. Tekinerdogan and H. Sozer. Defining architectural viewpoints for quality concerns. In *Proceedings of the 5th European Conference on Software Architecture*, pages 26–34, Essen, Germany, 2011.

[30] S. Ciraci, H. Sozer, M. Aksit, and W. Havinga. Guiding architects in selecting architectural evolution alternatives. In *Proceedings of the 5th European Conference on Software Architecture*, pages 252–260, Essen, Germany, 2011.

[31] S. Ciraci, H. Sozer, M. Aksit, and W. Havinga. Execution constraint verification of exception handling on uml sequence diagrams. In *Proceedings of the 5th IEEE International Conference on Secure Software Integration and Reliability Improvement*, pages 31–40, Jeju Island, Korea, 2011.

[32] A. de Roo, H. Sozer, and M. Aksit. Runtime verification of domain-specific models of physical characteristics in control software. In *Proceedings of the 5th IEEE International Conference on Secure Software Integration and Reliability Improvement*, pages 41–50, Jeju Island, Korea, 2011.

[33] K. Hatun, C. Bockisch, H. Sozer, and M. Aksit. A feature model and development approach for schedulers. In *Proceedings of the 1st Workshop on Modularity in Systems Software*, pages 1–5, Porto de Galinhas, Brazil, 2011.

[34] H. Sozer, R. Abreu, M. Aksit, and A. J.C. van Gemund. Increasing system availability with local recovery based on fault localization. In *Proceedings of the 10th International Conference on Quality Software*, Zhangjiajie, China, 2010.

[35] A. de Roo, H. Sozer, and M. Aksit. An architectural style for optimizing system qualities in adaptive embedded systems using multi-objective optimization. In *Proceedings of the 8th Working IEEE/IFIP Conference on Software Architecture*, pages 349–352, Cambridge, UK, 2009.

[36] H. Boudali and H. Sozer and M. Stoelinga. Architectural availability analysis of software decomposition for local recovery. In *Proceedings of the 3rd IEEE International Conference on Secure Software Integration and Reliability Improvement*, pages 14–22, Shanghai, China, 2009.

[37] H. Sozer and B. Tekinerdogan. Introducing recovery style for modeling and analyzing system recovery. In *Proceedings of the 7th Working IEEE/IFIP Conference on Software Architecture*, pages 167–176, Vancouver, Canada, 2008.

[38] H. Sozer, C. Hofmann, B. Tekinerdogan, and M. Aksit. Detecting mode inconsistencies in component-based embedded software. In *Proceedings of the Workshop on Architecting Dependable Systems, International Conference on Dependable Systems*, Edinburgh, UK, 2007.

[39] B. Tekinerdogan, H. Sozer, and M. Aksit. Software architecture reliability analysis using failure scenarios. In *Proceedings of the 7th Working IEEE/IFIP Conference on Software Architecture*, pages 203–204, Pittsburgh, US, 2005.

[40] M. Tekkalmaz, H. Sozer, and I. Korpeoglu. Distributed construction and maintenance of bandwidth-efficient bluetooth scatternets. In *Proceedings of the IEEE International Conference on Communications*, pages 3223–3229, Seoul, South Korea, 2005.

[41] H. Sozer, M. Tekkalmaz, and I. Korpeoglu. A peer-to-peer file sharing system for wireless ad-hoc networks. In *Proceedings of the Third Annual Mediterranean Ad Hoc Networking Workshop*, Bodrum, Turkey, 2004.

Publications in National Conferences

[1] Omer Faruk Erdil, Irfan Can, and Hasan Sozer. Test kahini olarak görüntü karşılaştırma algoritmalarının değerlendirilmesi. In *Proceedings of the 11th Turkish National Software Engineering Symposium*, pages 101–113, Alanya, Turkey, 2017.

[2] Ersin Ersoy, Metin Altinisik, and Hasan Sozer. PL/SQL programları İçin veri tabanı bağımlılık analizi. In *Proceedings of the 11th Turkish National Software Engineering Symposium*, pages 114–125, Alanya, Turkey, 2017.

[3] C. Sahin Gebizli, D. Metin, and H. Sozer. Statik ve dinamik analizler ile hesaplanan risklere dayalı olarak test modellerinin İyileştirilmesi. In *Proceedings of the 9th Turkish National Software Engineering Symposium*, pages 283–293, Izmir, Turkey, 2015. (best paper award).

[4] U. Ozkan and H.Sozer. Web uygulamaları için model bazlı test süreci otomasyonu. In *Proceedings of the 9th Turkish National Software Engineering Symposium*, pages 101–112, Izmir, Turkey, 2015.

[5] D. Kurt and H. Sozer. Geçmişe dönük hata tespit oranlarının zamanla değişen etkisine göre regresyon testlerinin Önceliklendirilmesi. In *Proceedings of the 8th Turkish National Software Engineering Symposium*, pages 281–292, Guzelyurt, Northern Cyprus, 2014.

[6] I. Ari, E. Olmezogullari, and H. Sozer. Application of combinatorial testing techniques in complex event processing engines. In *Proceedings of the 20th Signal Processing and Communications Applications Conference*, pages 1–4, Mugla, Turkey, 2012.

Theses

[1] H. Sozer. *Architecting Fault-Tolerant Software Systems*. PhD thesis, University of Twente, Enschede, The Netherlands, January 2009.

[2] H. Sozer. A peer-to-peer file sharing system for wireless ad-hoc networks. Master's thesis, Bilkent University, Ankara, Turkey, August 2004.

Teaching

Courses Taught

Undergraduate	Software Engineering
	Computer Programming
	Data Structures and Algorithms
Graduate	Agile Software Development
	Software Testing and Analysis
	Advanced Software Engineering
	Design of Software Architectures
	Software Product Line Engineering

Supervised Ph.D. thesis

2017 Ceren Şahin Gebizli, *Automated Refinement of Models for Model-Based Testing*, Ozyegin University, İstanbul, Turkey

2012 Arjan de Roo, *Managing Software Complexity of Adaptive Systems*, University of Twente, Enschede, The Netherlands, co-supervised with Dr. Lodewijk Bergmans and Dr. Mehmet Akşit

Supervised M.Sc. thesis

2018 Abdulhadi Kırkıcı, *Risk-Driven Model-Based Testing*, Ozyegin University, İstanbul, Turkey

2017 Burcu Özten Ergun, *Tool Support for Model Based Software Product Line Testing*, Ozyegin University, İstanbul, Turkey

2017 Yunus Kılıç, *Generating Runtime Verification Specifications Based on Static Code Analysis Alerts*, Ozyegin University, İstanbul, Turkey

2016 Ersin Ersoy, *Extending Static Analysis with Application-Specific Rules by Analyzing Runtime Execution Traces*, Ozyegin University, İstanbul, Turkey

2016 Metin Altınışık, *Automated Procedure Clustering for Reverse Engineering PL/SQL Programs*, Ozyegin University, İstanbul, Turkey, co-supervised with Dr. Gonca Gürsun

2015 Gün Karagöz, *Reproducing Field Failures based on Semi-formal Failure Scenario Descriptions*, Ozyegin University, İstanbul, Turkey

2014 Arda Ahmet Ünsal, *Domain-Specific Adaptive Service Monitoring*, Ozyegin University, İstanbul, Turkey, co-supervised with Dr. Tankut Barış Aktemur

2014 Ceren Şahin Gebizli, *Improving Models for Model-based Testing based on Exploratory Testing*, Ozyegin University, İstanbul, Turkey

2013 Koray Gülcü, *Fault Masking as a Service*, Ozyegin University, İstanbul, Turkey, co-supervised with Dr. Tankut Barış Aktemur

2012 Merve Astekin, *Code Clones in Simulation Software Systems*, İstanbul Technical University, İstanbul, Turkey, co-supervised with Dr. Muhittin Gökmen

2011 Sander Nouta, *Data processing networks made easy*, University of Twente, Enschede, The Netherlands, co-supervised with Dr. Andreas Wombacher

2010 Ronald Peterson, *Testing real-time requirements for integrated systems*, University of Twente, Enschede, The Netherlands, co-supervised with Dr. Ander de Keijzer

2008 Wouter Roosenburg, *Design of a reasoning framework for simulation-based performance analysis*, University of Twente, Enschede, The Netherlands, co-supervised with Dr. Bedir Tekinerdoğan and Thales Nederland

Academic Service

Reviewing
Information and Software Technology
Journal of Systems and Software
Software: Practice and Experience
IEEE Transactions on Reliability
Turkish Journal of Electrical Engineering and Computer Sciences
International Journal of Software Engineering and Knowledge Engineering
Computer Communications
ACM Computing Reviews

PC membership
ICTSS'18, SNPD'18, ACM SAC'14-18 (Track on Software Engineering),
TAIC PART'17-18, UYMS'09-18, ECSA'16, WICSA'16,
ACM SAC'14-15 (Track on Software Verification and Testing), UYMK'10-18

Organization
ECSA'17 (Publicity co-chair)

Awards & Honours

2015 **Best Paper Award**.
Turkish National Software Engineering Symposium (UYMS 2015)

2014 **National Young Researcher Career Award**.
The Scientific and Technical Research Council of Turkey (TUBITAK)

2002 – 2004 **High Honour Success scholarship**.
M.Sc. studies at Bilkent University, Ankara, Turkey

2000 – 2002 **High Honour Success scholarship**.
B.Sc. studies at Bilkent University, Ankara, Turkey

Languages

native **Turkish**

fluent **English**

intermediate **Dutch**